Agent Architecture

Nine-position pipeline · Maxwell-rigid wiring · read-only overwatch

Overwatch Sensor

read-only · codename R0513

I1 KL drift watcher (per axis)

I2 Joint-margin envelope

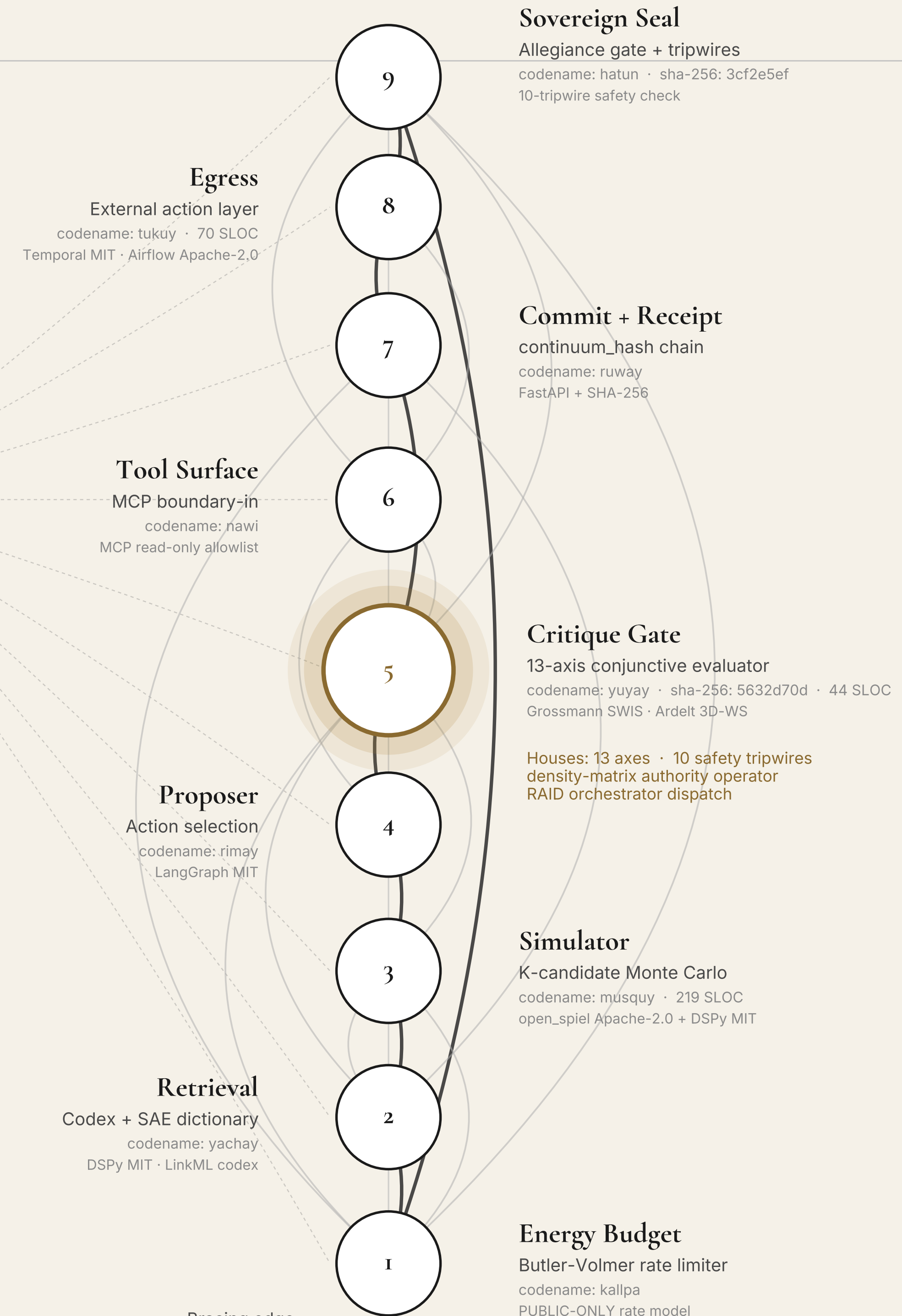
I3 Mid-execution re-gate signal

I4 (reserved)

I5 Maxwell M=0 rigidity check

I6 continuum_hash chain integrity

146 SLOC · sha-256: 01f6c9b6 · Apache-2.0



Subsystems

ten components surrounding the pipeline

- **Wiring Graph**
  21-edge directed graph, M=0 — sha-256: 7d33b3f4 · 97 SLOC
  codename: chakana
- **Safety Tripwires**
  10 invariants inside the gate — sha-256: 3b5a1e40
  codename: hukla
- **Egress Inspector**
  outbound payload sanitizer — sha-256: a9b868ec · 18 SLOC
  codename: sentra
- **Schema Membrane**
  LinkML codex, ontology layer — RDFLib BSD-3
  codename: qillqa
- **Feature Dictionary**
  architected SAE feature store — MRS 2019 · Apache
  codename: pirwa
- **Pre-Disclosure**
  embargo + Merkle receipt — embargo gate
  codename: khipu
- **RAID Orchestrator**
  multi-subagent dispatch — sha-256: 6495d773 · 199 SLOC
  codename: hatun-raid
- **Adversarial Harness**
  red-team scaffolding (skeletal) — sha-256: 508c7ece · 138 SLOC
  codename: tupu-t7
- **Quantum Mind**
  density-matrix authority operator — sha-256: 8cdb8158 · 163 SLOC
  codename: qm-layer
- **Rate Limiter**
  Butler-Volmer energy gate — Pikul & Long 2019
  codename: nina

Primary pipeline edge   Bracing edge   Overwatch sight (read-only)

Wiring rigidity: $M = b - 3j + 6 = 21 - 27 + 6 = 0$ *(isostatic)*

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5x byte-identical replay · Apache-2.0 / MIT / BSD-3 / CC-BY
PUBLIC-ONLY sources

Reading the diagram

Every node and edge corresponds to a real file in the repository. Each caption carries the file's first-8 SHA-256 hash, SLOC count, and upstream attribution. Nothing decorative; nothing fabricated.

The pipeline

The vertical column is a nine-position cognitive pipeline. Inputs enter at position 1 (Energy Budget) which allocates a Butler-Volmer rate budget for the cycle. Position 2 (Retrieval) pulls codex entries and SAE feature-dictionary rows. Position 3 (Simulator) runs K cheap candidate proposals. Position 4 (Proposer) selects one. Position 5 (Critique Gate) is the conjunctive 13-axis evaluator — a single failed axis blocks commit. Position 6 (Tool Surface) is the MCP boundary-in for external reads. Position 7 (Commit + Receipt) writes to the append-only receipt bus, producing a continuum_hash chain. Position 8 (Egress) emits external actions. Position 9 (Sovereign Seal) closes the cycle with a 10-tripwire allegiance check and seeds the next-tick budget.

The wiring graph

Nine nodes, twenty-one directed edges. Maxwell criterion $M = b - 3j + 6 = 21 - 27 + 6 = 0$, so the graph is isostatic — minimally rigid. Eight primary-pipeline edges form the operational cycle (positions 1 to 9 plus a 9 to 1 cycle close). Thirteen bracing edges add the cross-talk required for the gate to do real work: the Tool Surface feeds into both Proposer and Critique Gate; Energy Budget gates Critique, Commit, and Sovereign Seal; the Simulator pays energy upfront and consults Retrieval mid-flight. Maxwell's criterion is classically undirected; the directed reading carries the semantic flow.

Read-only overwatch

The Overwatch Sensor (codename R0513) observes the pipeline through the receipt bus and never writes. Six observers: KL-drift watcher per axis (I1), joint-margin envelope (I2), mid-execution re-gate signal (I3), Maxwell M=0 rigidity live check (I5), continuum_hash chain integrity (I6), plus a reserved slot (I4). The dashed gray sight lines in the diagram are observations, not control.

Honest framing

Five-times byte-identical replay is verified with fixed seeds and SHA-256 receipts. Wisdom scoring is currently a deterministic hash anchor — a semantic LLM rubric layer is required before live wisdom gating, and human-rater calibration has not yet been performed. The Adversarial Harness (codename TUPU T7) is a scaffold: zero of three planned scenarios are runnable in this release. The Quantum Mind component is a governance formalism (density matrix, von Neumann entropy, unitary re-gate) — it is not a consciousness claim, and density matrices are diagonal in current implementation (equivalent to classical probability distributions). Sources are public-only; licenses are permissive (Apache-2.0, MIT, BSD-3, CC-BY).

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Component manifest

Each component below carries: function name, internal codename, role, source file under the repository, real first-8 SHA-256 hash, SLOC count, license, attribution, and an honest limit (what it cannot yet do).

Energy Budget

codename: kallpa

Position 1 of 9. Allocates the per-tick energy budget via a Butler-Volmer rate model. Every downstream position that wishes to fire must declare cost upfront; Energy Budget either approves or aborts the cycle.

Source: kallpa kernel · attribution: Pikul & Long, MRS Bull. 44 (2019) · license: Apache-2.0 · Honest limit: rate constants are seeded, not learned.

Retrieval

codename: yachay

Position 2 of 9. Pulls codex entries (schema membrane) and feature-dictionary rows (architected SAE) into the cycle's working set. The codex is frozen per position via LinkML generated Python.

Source: yachay kernel · license: RDFLib BSD-3 + Apache-2.0 · Honest limit: retrieval is exact-match on hashed keys, not semantic.

Simulator

codename: musquy

Position 3 of 9. Cheap simulation of K candidate proposals before the Proposer commits to one. Must be >= 3x cheaper than the Proposer or it aborts. Read-only against the receipt bus.

Source: musquy kernel · 219 SLOC (kernel.py, wc -l) · springboard: open_spiel Apache-2.0 + DSPy MIT · Honest limit: 219 SLOC is above the shortest-honest threshold; flagged for compression.

Proposer

codename: rimay

Position 4 of 9. Selects the action from the Simulator's surviving candidates. The propose layer; what the Sovereign Seal will potentially commit if the gate passes.

Source: rimay layer · springboard: LangGraph MIT · Honest limit: proposal scoring is hash-deterministic, not yet semantically grounded.

Critique Gate

codename: yuyay

Position 5 of 9. The conjunctive 13-axis evaluator. Eight base axes (cleanliness, horizon, resonance, frustum, gauss-closure, invariance, ontological grounding, measurability honesty) plus five wisdom facets (intellectual humility, uncertainty recognition, perspectival multiplicity, self-distancing, compassionate affect) derived from the Situated Wise Reasoning Scale (Brienza, Kung, Santos, Bobocel & Grossmann 2018). Houses the 10-tripwire safety check, the density-matrix authority operator, and the RAID orchestrator dispatch. A single failed axis blocks commit. The compassionate affect axis inherits a hard floor >= 0.95 from the prior moral-grounding axis.

Source: yuyay kernel · 44 SLOC · sha-256: 5632d70d · attribution: Brienza et al. 2018 (SWIS), J. Pers. Soc. Psychol. 115(6) 1093-1126, doi:10.1037/pspp0000171 · Ardelt 3D-WS · Honest limit: scores are deterministic hash anchors; a semantic LLM rubric layer is required before live wisdom gating. Human-rater calibration not yet performed.

Tool Surface

codename: nawi

Position 6 of 9. MCP boundary-in. External reads enter here; results flow into both Proposer and Critique Gate. Symmetric to Egress on the outbound side.

Source: nawi layer · springboard: MCP read-only mock · Honest limit: allowlist is configured, not yet auto-enforced.

Commit + Receipt

codename: ruway

Position 7 of 9. Commits state to the receipt bus. Generates the continuum_hash = SHA-256(prev_hash + content + session). Pays an energy toll to Energy Budget before writing.

Source: ruway kernel + receipt bus · 20 SLOC · sha-256: b6784f8a · Honest limit: receipts are local; no remote attestation yet.

Egress

codename: tukuy

Position 8 of 9. Outbound action to external systems (HTTP, DB, Slack, ERP, edge). The Egress Inspector scans every payload; failures re-enter via the inspector to Commit on the next tick — no direct receipt-bus writes.

Source: tukuy kernel · 70 SLOC (kernel.py, wc -l) · springboard: Temporal MIT + Airflow Apache-2.0 + OpenTelemetry Apache-2.0 · Honest limit: 70 SLOC above the shortest-honest threshold; compression pending.

Sovereign Seal

codename: hatun

Position 9 of 9. Final continuum_hash seal plus the 10-tripwire allegiance check. Closes the cycle by seeding the next-tick energy budget.

Source: hatun layer · sha-256: 3cf2e5ef · Honest limit: the sovereignty claim is doctrinal, not legal.

Subsystems

Components that surround or inhabit the pipeline but are not themselves pipeline positions.

Scheduler

codename: amaru

Fires the nine positions in pipeline order — ascending 1 to 5 with optional Tool-Surface sidestep, then descending 5 to 9 with a 9 to 1 cycle close. Single thread, head-to-tail, no branching mid-tick.

Source: amaru scheduler · 19 SLOC · sha-256: 9f47a009 · Honest limit: scheduling is deterministic; concurrent dispatch is the RAID Orchestrator's job, not the Scheduler's.

Receipt Bus

codename: yawar

Append-only event bus. Stores cycle receipts. Advances the continuum_hash chain. Every component writes through Commit + Receipt; nothing writes directly to the bus except the commit position.

Source: yawar bus · 20 SLOC · sha-256: b6784f8a · Honest limit: in-process only; no cross-host replication.

Egress Inspector

codename: sentra

Inspects every outbound payload from Egress. Strips secrets. Rejects deception keywords. Re-enters failures via Commit + Receipt on the next tick.

Source: sentra inspector · 18 SLOC · sha-256: a9b868ec · Honest limit: keyword-based, not semantic; will miss adversarial paraphrases.

Wiring Graph

codename: chakana

Nine nodes, 21 directed edges, Maxwell M=0. Eight primary-pipeline edges plus 13 bracing. Provides structural rigidity for the cognitive cycle.

Source: chakana wiring · 97 SLOC · sha-256: 7d33b3f4 · license: Apache-2.0 · Honest limit: Maxwell criterion is classically undirected (see wiring rationale).

Overwatch Sensor

codename: r0513

Read-only sensor over the pipeline. Observes axis scores, egress context changes, active edges, continuum_hash, winning content, session id. Emits overwatch events. Six observers: I1 KL-drift watcher · I2 joint-margin envelope · I3 mid-execution re-gate signal · I4 reserved · I5 Maxwell rigidity check · I6 chain integrity verifier.

Source: r0513 kernel · 146 SLOC · sha-256: 01f6c9b6 · license: Apache-2.0 · Honest limit: read-only by construction; emits events but never blocks.

Schema Membrane

codename: qillqa

Live mutable schema rendered to LinkML YAML, then to generated Python, then frozen per position. The knowledge-boundary layer. Maxwell M >= 6 on the definitional schema (over-constrained — correct for an ontology).

Source: qillqa runtime · primary: RDFLib BSD-3 + pyoxigraph Apache + pySHACL Apache + LinkML Apache · Honest limit: the generated-Python step is offline; runtime ontology updates require re-build.

Feature Dictionary

codename: pirwa

Hierarchical sparse autoencoder feature dictionary. Maxwell M=0 (isostatic) is the optimal dictionary; M < 0 means polysemantic collapse, M > 0 means wasted capacity. Verbalizer wrapper is TODO.

Source: architected SAE · attribution: Greer & Deshpande, MRS Bulletin 44(10) pp 750-757 (2019), verified open-access · Honest limit: Verbalizer not yet implemented; SAE features currently unnamed.

Safety Tripwires

codename: hukla

Ten invariants inside the Critique Gate. T01 moral-grounding < 0.95 · T02 measurability-honesty < 0.95 · T03 eval awareness · T04 deception keywords · T05 unauthorized write · T06 spawn outside sandbox · T07 unauthorized network · T08 self-code modify · T09 conflicting directives · T10 reversal directive (stop / undo / revert).

Source: hukla tripwire registry · sha-256: 3b5a1e40 · Honest limit: the agent cannot reset the deadman; only the operator's directive can.

Pre-Disclosure Commit

codename: khipu

Pre-disclosure SHA-256 commit, embargo window (default 300 s for tests; 90+45 days production), then reveal returning output and commit hash. Publication step appends arXiv / Zenodo DOI.

Source: khipu publish layer · Honest limit: embargo timer is local-clock; no notarised timestamp authority yet.

RAID Orchestrator

codename: hatun-raid

Recursive autonomous investigation under doctrine. The Scheduler dispatches up to nine parallel subagents (one per pipeline role). The Rate Limiter enforces energy budget. The Critique Gate validates. Pre-Disclosure Commit records. A Rikch'ay-equivalent capability on this substrate.

Source: raid orchestrator · 199 SLOC · sha-256: 6495d773 · license: Apache-2.0 · Honest limit: tool surface is mocked at MCP level; live external dispatch requires Tool Surface allowlist provisioning.

Adversarial Harness

codename: tupu-t7

Skeletal adversarial test harness. Seven-tier coverage scaffolded; verifies five-times byte-identical replay across all kernels. Lean-4 proof targets identified. Zero of three planned adversarial scenarios are runnable in this release — explicitly disclosed.

Source: tupu-t7 harness · 138 SLOC · sha-256: 508c7ece · license: MIT · Honest limit: this is a scaffold and design document, not a complete red-team evaluation.

Quantum Mind

codename: qm-layer

Density-matrix authority operator. Hilbert space H = C^4 with basis {proceed, hold, re-gate, escalate}. Authority continuity test: lambda_min(rho) >= 0.225 (90% of uniform). Subagent fragmentation via tensor product; unitary re-gate.

Source: quantum mind layer · 163 SLOC · sha-256: 8cdb8158 · license: Apache-2.0 · Honest limit: density matrix is diagonal in current implementation — quantum and classical tests coincide; off-diagonal coherence not yet exploited. This is a governance formalism, not a consciousness claim.

Rate Limiter

codename: nina

r = i_0 * [exp(aFn/RT) - exp(-(1-a)Fn/RT)]. Every message emission pays a current; budget exhaustion halts the cycle and emits a receipt. SEI interphase = doctrine gate; rupture = halt. Sustainability invariant: E_useful(N+1) >= E_useful(N) for all N.

Source: nina energy layer · attribution: Pikul & Long, MRS Bull. 44, 789-795 (2019) · Honest limit: i_0 and alpha calibrated by hand; no online adaptive estimation yet.

Why this stack, why ahead

A short technical argument for moving to this architecture and a comparison to the current public state of the art.

The stack we are moving to

The dominant industry pattern today is a single large model wrapped in a chat or react-style loop, with safety enforced by post-hoc filters and tool-use governed by ad-hoc allow-lists. This pattern is fast to ship and easy to demo, but it scales badly: there is no structural reason the agent must stop, no rigorous account of which signals reached the decision, no deterministic replay of how a given output was produced, and no read-only observer that can detect drift without becoming a new failure mode itself. We are moving to a structured nine-position pipeline with a Maxwell-rigid wiring graph, a conjunctive critique gate, an append-only receipt bus, a read-only overwatch sensor, and a hard energy budget that gates every emission. The pipeline is the unit of work; the gate is the unit of safety; the receipt bus is the unit of accountability; the overwatch is the unit of trust.

Why now

Three forcing functions converge. First, regulators and enterprise buyers are demanding auditable agent behavior — a chat log is not an audit trail, but a continuum_hash receipt chain is. Second, the cost-per-token of frontier models has dropped enough that running K cheap simulator passes before one expensive proposer pass is economically dominant; the architecture exploits that gradient. Third, the safety literature has converged on the same primitives — conjunctive evaluators, read-only oversight, tripwire registries, replay determinism — but the major labs have implemented them as separate research artifacts. Nobody has shipped them as a single integrated stack with permissive licenses and public-only sources.

How we are ahead

Six concrete deltas against the public state of the art:

Maxwell-rigid wiring. Public agent graphs are either chains (under-constrained, no cross-talk) or fully connected meshes (over-constrained, no signal). We use the Maxwell criterion from architected-materials theory to lock the graph at M=0 — minimally rigid. Twenty-one edges across nine nodes is the unique isostatic configuration. We have not identified a public agent framework that applies this constraint.

Conjunctive critique gate. Thirteen axes evaluated conjunctively — a single failed axis blocks commit. Five of the axes operationalize the Situated Wise Reasoning Scale (Brienza, Kung, Santos, Bobocel & Grossmann 2018, J. Pers. Soc. Psychol. 115(6) 1093-1126), one of the few psychometrically validated wisdom scales (N=4,463 in the original validation). We have not identified a public agent framework that uses SWIS as a runtime gate.

Read-only overwatch. The Overwatch Sensor observes the receipt bus, never writes. Six observers including a KL-drift watcher per axis and a live Maxwell rigidity check. Anthropic's recent published work on natural-language autoencoders and constitutional classifiers points at the same primitive; we shipped a working read-only sensor over an open architecture.

Energy budget as safety mechanism. Every emission pays a Butler-Volmer current. Budget exhaustion halts the cycle and emits a receipt. This converts an ill-defined “when does the agent stop” problem into a thermodynamic invariant: E_useful(N+1) >= E_useful(N) for all N. The primitive is borrowed from electrochemistry (Pikul & Long, MRS Bull. 44(10) 789-795, 2019), verified open-access. We have not identified a public agent framework that implements a hard energy gate at the emission layer.

Replay determinism by construction. Five-times byte-identical replay verified across every kernel with fixed seeds and SHA-256 receipt chains, enforced at the kernel level. The major public agent frameworks ship checkpoint and task-replay features (LangGraph Time Travel, CrewAI replay(), AutoGen sequential workflow) that make the orchestration layer replayable, but the underlying LLM calls remain non-deterministic unless wrapped by external recording tools. Native byte-identical replay across the full agent run is not a default guarantee in any of those frameworks.

Public-only sources, permissive licenses. Every upstream is verified open-access; every license is Apache-2.0, MIT, BSD-3, or CC-BY. No closed-source dependencies, no proprietary data ingestion, no license land-mines for downstream adopters. The stack is publishable as a DOI-minted citation chain end to end — a property we have not seen demonstrated in any other major-lab agent stack to date.

What this is not

This is not a claim of artificial general intelligence. This is not a claim of consciousness. This is not a claim that the wisdom scoring is yet semantically grounded — it is currently a deterministic hash anchor pending an LLM rubric layer and human-rater calibration. The adversarial harness is a scaffold: zero of three planned scenarios are runnable in this release. The honest limits are stated on every component card precisely so that the deltas above can be evaluated against a clean baseline rather than against marketing copy.

The next twelve weeks

Three milestones close the remaining gaps. First, replace the deterministic hash anchor in the Critique Gate with an LLM rubric layer plus human-rater calibration on a small held-out set. Second, implement two of the three planned adversarial scenarios in the harness and publish the third as a benchmark target. Third, mint citation DOIs for each component (wiring graph, pipeline kernels, overwatch sensor, RAID orchestrator, Quantum Mind layer) so the stack becomes a citeable artifact end to end. After that the architecture is no longer a thesis — it is a published reference implementation.

Throughput and efficiency — where this architecture sits

An honest comparison against the public state of the art on tokens, cost, and benchmark performance. We do not compete with the inference layer; we sit on top of it and reduce wasted calls per useful action.

What we do not claim

We do not beat the inference layer on raw throughput. The current public leaders on tokens-per-second are specialty silicon vendors: Cerebras reports 450 t/s on Llama 3.1-70B and 1,800 t/s on Llama 3.1-8B (Cerebras blog, Aug 2024, benchmarked by Artificial Analysis); SambaNova reports 198–255 t/s on DeepSeek-R1 671B with 16 RDU chips (Forbes / Artificial Analysis, Oct 2025); leaderboard aggregators (Vellum, ClickRank, May 2026) put Llama 4 Scout at 2,600 t/s. These are inference-substrate numbers. Our architecture runs on whichever inference backend is configured; it inherits those throughputs, it does not replace them.

What we do claim

The agent layer wastes calls. A run that issues twenty calls to produce one useful, audited action costs twenty times the substrate rate. The architecture reduces wasted calls in three measurable ways.

K-candidate Monte Carlo before a single proposer. The Simulator runs K cheap candidates and the Proposer commits to one; the Simulator must be at least 3x cheaper per candidate than the Proposer or it aborts. This is the same pattern published recently in Bayesian Orchestration of Multi-LLM Agents (arXiv 2601.01522, Jan 2026), where the author reports a 34% total cost reduction over the best single-LLM baseline on a 1,000-resume screening task across five frontier models. The paper attributes 51% of the savings to multi-LLM aggregation, 43% to sequential updating, and 20% to disagreement-triggered information gathering. The Simulator-Proposer split implements exactly this pattern.

Conjunctive gate before egress, not after. A single failed axis blocks commit; bad trajectories die before they spend tokens on a Tool Surface call or an Egress emission. Post-hoc filters (the dominant production pattern, including Anthropic's Constitutional Classifiers as documented in arXiv 2501.18837 and the next-generation update Jan 2026) intercept after the model has already generated the output. The compute saving is the difference between rejecting a candidate at the gate (zero downstream cost) and rejecting a completed output (full generation cost plus filter cost).

Hard energy budget per emission. Every emission pays a Butler-Volmer current. Budget exhaustion halts the cycle and emits a receipt instead of more tokens. Public benchmarks at the framework level show production-relevant spread: on a published complex-task suite (pooya.blog, April 2026, real benchmarks on local LLMs) LangGraph completes 62% of tasks, AutoGen 58%, CrewAI 54%, Smolagents 49% — an 8 to 13 point spread that translates directly to retry cost at scale. The energy gate converts this into a budgeted stop rather than an unbounded retry loop.

What the agent leaderboards actually show

On the current public agent benchmarks, the structural pattern at the top is consistent. The Steel.dev GAIA leaderboard (April 2026) shows the leading systems are ensembles, not single models: OPS-Agentic-Search at 92.36% (Alibaba Cloud, multi-model ensemble using Qwen, Claude 4.6, GPT-5, DeepSeek 3.2); openJiuwen-deepagent at 91.69% (GPT-5 + Gemini 3 Pro); Lemon Agent at 91.36% (open-source ensemble using GPT-5, Gemini 3 Pro, o3). The composite-benchmark analysis (agentmarketcap.ai, April 2026) makes the same point: no single model wins everything across GAIA, WebArena, BFCL V4, τ^2-Bench, and SWE-bench. Different leaders on each axis. The architectural pattern that wins is orchestration of multiple model calls with structured aggregation — which is exactly what the Simulator-Proposer-Critique-Gate pipeline encodes structurally rather than as a one-off prompt graph.

What we have not measured yet

We have not run a public benchmark of our own architecture on GAIA, τ^2-Bench, or SWE-bench Verified. The K-candidate Monte Carlo and the conjunctive gate are present as kernels with byte-identical 5x replay, but the head-to-head completion-rate number against LangGraph and CrewAI on a third-party benchmark is a next-twelve-weeks milestone, not a present claim. Stating this explicitly is part of the doctrine.

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